

DSAA3073: Theories in Data Science

Week 3 Validation Sheet · From Empirical Risk to Finite-Sample Complexity

Coverage: Week 3

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. On a sample S , a classifier h_1 makes 3 mistakes out of 12 and h_2 makes 2 mistakes out of 12. Which conclusion follows from these facts alone?

- (a) $R(h_2) = \frac{1}{6}$.
- (b) h_2 is an empirical risk minimizer within $\{h_1, h_2\}$.
- (c) h_2 has smaller population risk than h_1 .
- (d) The distribution is realizable by $\{h_1, h_2\}$.

A2 [5 marks]. Which observation is sufficient to establish that a distribution is realizable by a binary class $\mathcal{H}$?

- (a) Some $h \in \mathcal{H}$ has zero empirical risk on one observed sample.
- (b) ERM returns a unique hypothesis.
- (c) Some $h^* \in \mathcal{H}$ has population risk $R(h^*) = 0$.
- (d) The sample size exceeds the number of hypotheses.

A3 [5 marks]. On one three-point set C , a binary class realizes exactly seven distinct label vectors. Which statement is necessarily correct?

- (a) C is shattered.
- (b) $\Pi_{\mathcal{H}}(3) = 7$.
- (c) $\Pi_{\mathcal{H}}(3) \geq 7$, but C is not shattered.
- (d) $\text{VCdim}(\mathcal{H}) = 3$.

A4 [5 marks]. Suppose $\text{VCdim}(\mathcal{H}) \leq 2$. What does Sauer–Shelah guarantee at $m = 5$?

- (a) $\Pi_{\mathcal{H}}(5) = 10$.
- (b) $\Pi_{\mathcal{H}}(5) \leq 1 + 5 + 10 = 16$.
- (c) $|\mathcal{H}| \leq 16$.
- (d) Every five-point set is shattered.

Part B: Definition in use

B1 [20 marks]. A finite class $\mathcal{H} = \{h_1, \dots, h_K\}$ is evaluated on the same i.i.d. sample S . For every index j fixed before observing S , bounded-loss concentration gives

$$P\left(\left|\hat{R}_{S(h_j)} - R(h_j)\right| > \gamma\right) \leq 2 \exp(-2m\gamma^2).$$

After seeing all empirical risks, ERM returns $h_{J(S)}$. An analyst simply replaces the fixed j by $J(S)$ in the displayed inequality. Audit that step, and derive a correct confidence- $1 - \delta$ statement that protects whichever hypothesis ERM selects. Give a sufficient sample-size condition.

Answer space

Part C: Basic skills

C1 [7 marks]. A finite binary class has $|\mathcal{H}| = 100$. Assume realizability and that ERM returns a consistent hypothesis. Compute a sufficient sample size for $\varepsilon = 0.20$ and $\delta = 0.10$ using

$$m \geq \left(\frac{1}{\varepsilon}\right) \left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)\right).$$

Round upward.

Answer space

C2 [8 marks]. On $C = \{x_1, x_2, x_3\}$, five hypotheses induce the vectors

000, 001, 001, 011, 111.

Compute $|\mathcal{H}|_C$, give the conclusion about $\Pi_{\mathcal{H}}(3)$, and decide whether C is shattered.

Answer space

Part D: Construction and obstruction

D1 [17 marks]. Let $\mathcal{T} = \{h_t : t \in \mathbb{R}\}$ be the threshold class $h_{t(x)} = \mathbf{1}_{\{x \geq t\}}$ on $\mathbb{R}$. Determine $\text{VCdim}(\mathcal{T})$. Your proof must include a concrete shattered set for the lower bound and an obstruction that works for every set of the next larger size.

Answer space

D2 [18 marks]. Let $\mathcal{I} = \{h_{a,b} : a \leq b\}$ be the interval class $h_{a,b}(x) = \mathbf{1}_{\{a \leq x \leq b\}}$ on $\mathbb{R}$. Establish its exact VC dimension. For the lower bound, use the set $\{-2, 3\}$ and give an explicit interval for each of its four target labelings. For the upper bound, your obstruction must apply to an arbitrary three-point set.

Answer space

Part E: Transfer to a new ordered class

E1 [25 marks]. Define a binary class $\mathcal{P}$ on $\mathbb{R}$ as follows: the positive region of a classifier is either a left ray $(-\infty, t]$ or a right ray $[t, \infty)$, for some $t \in \mathbb{R}$. Thus each classifier labels either an initial portion or a final portion of the real line as 1.

Determine the exact growth function $\Pi_{\mathcal{P}}(m)$ for every integer $m \geq 1$ and the exact value of $\text{VCdim}(\mathcal{P})$. Give a complete proof of both claims; a count from one specially chosen set alone does not establish the growth function or the VC upper bound.

Answer space

Part F: Compare two valid finite-class routes

F1 [15 marks]. Let $|\mathcal{H}| = 64$ and $\delta = 0.05$. Compare the following two requests.

- Request S asks for one event on which every $h \in \mathcal{H}$ satisfies $|\hat{R}_{S(h)} - R(h)| \leq \gamma$ with $\gamma = 0.10$.
- Request R assumes binary zero-one loss, realizability, and a consistent ERM, and asks that its returned hypothesis have risk at most $\varepsilon = 0.10$.

Compute a sufficient sample size for each request using the appropriate Week 3 finite-class route. Then explain why the smaller number cannot replace the larger one when Request S is required.

Answer space